

Delhi Public School

Subject: Physics | Grade: 12th | Class: 12th

Time: 1 Hour

Maximum Marks: 30

Name: _____

Roll No: _____

Section: _____

General Instructions:

1. Read all questions carefully before answering.
2. Write neatly and legibly.

Section A

This section contains multiple-choice questions, true/false assertions, and fill-in-the-blank questions. Each question carries 1 mark.

Q1. What is the electric potential due to a point charge 'q' at a distance 'r' from it in vacuum? [Easy] [1]

- a. $q / (4 \pi \epsilon_0 r^2)$
- b. $q / (4 \pi \epsilon_0 r)$
- c. $q^2 / (4 \pi \epsilon_0 r)$
- d. $q / (4 \pi \epsilon_0 r^3)$

Q2. If the current flowing through a copper wire is doubled, what happens to the drift velocity of the free electrons inside the wire (assuming temperature remains constant)? [Moderate] [1]

- a. It remains unchanged.
- b. It becomes halved.
- c. It is doubled.
- d. It becomes four times.

Q3. Which of the following is the correct SI unit of magnetic dipole moment? [Moderate] [1]

- a. Ampere-meter
- b. Ampere-meter square
- c. Tesla-meter / Ampere
- d. Tesla-meter square

Q4. In a purely capacitive AC circuit, what is the phase relationship between the alternating voltage across the capacitor and the current passing through it? [Hard] [1]

- a. Voltage and current are in phase.
- b. Voltage leads current by $\pi / 2$ radians.
- c. Current leads voltage by $\pi / 2$ radians.
- d. Voltage leads current by π radians.

Q5. Which of the following expressions correctly represents the energy of a photon of wavelength λ ? [Easy] [1]

- a. $h \cdot c / \lambda$
- b. $h \cdot \lambda / c$
- c. $h \cdot c \cdot \lambda$
- d. $e \cdot V$

Q6. Electrostatic field lines always form continuous closed loops starting from positive charges and ending at negative charges. [Easy] [1]

Q7. The resistivity of a semiconductor material decreases with an increase in its temperature. [Moderate] [1]

Q8. According to Gauss's Law for Magnetism, the net magnetic flux through any closed surface is always zero. [Moderate] [1]

Q9. A step-up transformer can be used to increase the net electrical power output of an alternating current source. [Easy] [1]

Q10. According to the de Broglie hypothesis, a heavier particle will always have a longer de Broglie wavelength than a lighter particle moving with the same velocity. [Hard] [1]

Q11. The phenomenon of splitting of white light into its component colors when passing through a prism is called _____. [Easy] [1]

Q12. The focal length of a plane mirror is _____. [Moderate] [1]

Q13. When a p-n junction diode is forward biased, the width of the depletion layer _____. [Moderate] [1]

Q14. When a dielectric slab of dielectric constant K is completely inserted between the plates of an isolated charged capacitor, its capacitance _____. [Moderate] [1]

Q15. The minimum energy required by an electron to escape from a metal surface is called its _____. [Hard] [1]

Section B

This section contains short-answer questions. Each question carries 2 marks.

Q16. Define an equipotential surface. Write any two important properties of an equipotential surface. [Easy] [2]

Q17. State Ohm's law. Draw a representative graph showing the variation of current with voltage for (i) Ohmic conductors, and (ii) Non-ohmic devices (like a diode). [Moderate] [2]

Q18. Two point charges of +2 microcoulombs and -2 microcoulombs are kept separated by a distance of 10 cm in vacuum. Determine the net electric potential at the midpoint of the line joining these two charges. [Hard] [2]

Section C

This section contains long-answer questions. Each question carries 3 marks.

Q19. State Biot-Savart Law and write its mathematical expression in vector form. Use this law to derive an expression for the magnetic field at the center of a circular current-carrying loop of radius R . [Moderate] [3]

Q20. State Faraday's Laws of Electromagnetic Induction. A square loop of side 10 cm and resistance 0.5 ohm is placed vertically in an east-west plane. A uniform magnetic field of 0.10 T is set up across the plane in the north-east direction. If the magnetic field is decreased steadily to zero in 0.70 seconds, calculate the magnitude of the induced electromotive force (emf) and current during this time. [Hard] **[3]**

Q21. Draw a neat labeled ray diagram showing the image formation of a distant object by a refracting telescope in normal adjustment. Write the formulas for its magnifying power and the overall length of the telescope tube in this adjustment. [Moderate] **[3]**

— End of Question Paper —